

UNITED STATES DISTRICT COURT
NORTHERN DISTRICT OF CALIFORNIA
SAN JOSE DIVISION

LEGATO SYSTEMS, INC.

Plaintiff,

Vs.

NETWORK SPECIALISTS, INC.

Defendant.

Case No. CO3 02286 JW (EAI)

**Declaration of W. Anthony Mason in
Support of NSI's Responsive Claim
Construction Brief**

I, W. Anthony Mason, state and declare as follows:

1. I am over the age of 21 years and am fully able and competent to testify to the matters set forth in this declaration, and have personal knowledge of these matters.

2. I received a Bachelor's degree in mathematics with computer science concentration from the University of Chicago in 1987. I have been involved in the development of operating system software, including file system drivers and file system filter drivers for several different operating systems for the past seventeen years. I have been an architect for a commercially available file system filter driver software development kit, and a file system development kit for the Windows operating system. I have also served as lead designer and developer for various file system and file system filter driver development projects for Windows. Further, I have served as lead developer and instructor of courseware specific to Windows.

3. I have authored over 30 publications in the field of internal software design and development, including a number of articles concerning file systems and device drivers and related technology. I am also co-author of a book entitled Windows NT Device Driver Development published in 1999. I am a member of the ACM (Association for Computing Machinery) and USENIX advanced computing systems associations and a Microsoft Most Valuable Professional. Based on my education and over 17 years experience in internal software design and especially in operating system software, including file system drivers and file system filter drivers, I am fully knowledgeable with the concepts discussed in the specification and file histories in U.S. Patent Nos. 5,799,141 and 6,308,283 (the asserted patents) and the operation of the data protection software described therein.

4. A copy of my curriculum vitae is attached as Exhibit A to this declaration and includes additional details about my qualifications. My curriculum vitae includes a list of publications authored by me within the past 10 years. I have not testified as an expert at trial or by deposition within the preceding four years. My time spent on this matter is being billed at my standard consulting rate of either \$500 per hour or \$4000 per day.

5. I have reviewed the following materials to determine the proper and correct scope of claims 1-14 of the '141 patent and claims 1-16 of the '283 patent.

- The '141 patent (Exhibit B) and references cited therein
- The '283 patent (Exhibit C) and references cited therein
- The file histories of the '141 patent and '283 patents¹
- Excerpts from Windows NT Device Driver Development (1999) (Exhibit D)
- Excerpts from Webster's Third New Int'l Dictionary (1993) (Exhibit E)
- Excerpts from Microsoft Developers Network (MSDN) Library, Understanding Driver and Operating System Basics (2003) (Exhibit F)
- Excerpts from Dictionary of Computer and Internet Words (2001) (Exhibit G)
- Excerpts from Newton's Telecom Dictionary (2003) (Exhibit H)
- Excerpts from Newton's Telecom Dictionary (1995). (Exhibit I)

¹ I understand that the file histories of the asserted patents were attached as Exhibits 7 and 8 to Legato's Opening Claim Construction Brief

1 6. I have been asked to provide opinions as to what a person of ordinary skill in
2 the art at the time of the effective filing date of the patents would have understood various
3 terms and phrases in the claims of the '141 and '283 patents to mean. It is my understanding
4 that the effective filing date of the asserted patents is June 9, 1995. Therefore, when I refer
5 to what a person of ordinary skill in the art would have understood, my opinions are from the
6 perspective of what such a person in the art at the time of the effective filing date in 1995
7 would have understood. Because the asserted patents share essentially the same
8 specification, I refer only to the '141 patent for brevity. In addition, my opinions expressed
9 herein also, from time to time, refer to dictionary definitions and other resources in further
10 support of the constructions proposed herein. Although in some instances, the publication
11 dates of certain resources are after the effective filing date, in my opinion, the definitions
12 referred to herein are nonetheless consistent with what a person of ordinary skill in the art
13 would have understood the terms to mean in 1995 based on my actual experience in the art
14 prior to, during and since this time frame.

15 7. The basis for my opinions are set out either in this declaration or in the claim
16 chart which I understand is attached as Exhibit 2 to NSI's Responsive Brief being filed on
17 May 14, 2004 and these documents should be read in conjunction with one another. In
18 addition to the documents cited in my declaration and the accompanying claim chart, my
19 opinions are based on my years of technical knowledge and experience in the area of internal
20 software design and development and particular experience with file systems and device
21 drivers.

22 23 **The Person Of Ordinary Skill In The Art**

24 8. Based on my years of experience in the internal software design and
25 development and especially in file systems and device drivers, it is my opinion that a person
26 of ordinary skill in the art concerning the asserted patents in 1995 would be a person with at
27 least a Bachelors degree in Computer Science or Computer Engineering, with at least four
28

1 years experience in internal software design and operating systems, including relevant
2 experience in the field of file systems and device drivers.

3 9. In addition, a person of skill in the art would understand that prior to 1995
4 (which I understand is the year of the effective filing date of the patent) that various
5 techniques have been available to back-up data, and data protection systems which backed-
6 up data files to a remote location (either in real-time or periodically) so that lost data could be
7 recovered in the event of a disaster were known in the art more than a year prior to the
8 effective filing date of the asserted patents.

9
10 **Claim Construction Generally**

11 10. According to my understanding, patent claims are construed in light of the
12 claims, the specification and the prosecution history of the patent. It is my understanding
13 that the words of a claim are presumed to have their ordinary meaning and that these words
14 are interpreted as one of ordinary skill would have understood them at the time of the
15 effective filing date of the patent. It is my understanding that dictionaries, encyclopedias and
16 treatises may be used to assist in determining the ordinary meaning of claim terms. It is my
17 understanding that if the claim language is clear on its face, then the specification and file
18 history are examined for consistency and to determine whether the presumption of ordinary
19 and customary meaning is rebutted. It is further my understanding that if the claim language
20 is unclear, then the specification and file history are examined to help resolve this lack of
21 clarity.

22 11. It is my understanding that extrinsic evidence, which I understand is evidence
23 other than the claims, specification and prosecution history and which I understand may
24 include, for example, dictionary definitions, treatises, prior art, and expert testimony, may be
25 relevant to the interpretation of claim terms in cases where the proper interpretation is not
26 clear from the claim language, specification and file history, or to provide the meaning of
27 terms of art and may be used to assist the court in its understanding of the meaning of terms
28 from the perspective of a person of ordinary skill in the art.

12. I understand that a claim may be written in “means-plus-function” format where the claim language recites a means for performing a specified function without reciting sufficient structure to perform the recited function. I understand that a claim so drafted is construed to cover only the particular corresponding structure described in the specification and equivalents of that structure. I understand that the use of the word “means” in a patent claim gives rise to a presumption that the inventor used the term advisedly to invoke the claim construction rules for means-plus-function clauses, whereas when the word “means” is not used, then it is presumed that the claim limitation is not expressed as a means-plus-function element. I understand that determining whether either of these presumption is rebutted requires focus on whether the properly construed claim recites sufficient structure to support performing entirely the recited function of the claim element.

Construction of Disputed Terms in the ‘141 and ‘283 Patent Claims

“mirroring driver” and “mirroring driver means”

13. In my opinion, one of ordinary skill would have understood the term “mirroring driver” in the context of the asserted patents to mean:

software configured to execute in an operating system
input/output stack which intercepts file changes and determines
from a table which changes are to be backed-up.

14. The terms “mirroring driver” and “mirroring driver means” appear in each of the claims of the asserted patents as follows:

‘141 Patent - Claim 1	a mirroring driver that captures change information representing an individual change to a file from a selected subset of the one or more data files by the at least one application program
‘141 Patent - Claim 9	intercepting an operation on a selected data file residing in a file system performed by an application program having no data protection code and executing on a local computer system . . . wherein the step of intercepting is performed by a mirroring driver that is attached to a file system driver on the local computer system

1 2 3	'141 Patent - Claim 14	mirroring driver means associated with the local computer system <i>for intercepting an operation performed by the at least one application program on the at least one of the one or more data files</i>
4 5 6	'283 Patent - Claim 1	a mirroring driver that: <i>intercepts change information initiated by the at least one application program, the change information representing a change to a selected file; and compares information identifying the selected file with the configuration database to determine whether the configuration database specifies that the selected data file is to be mirrored</i>
7 8 9 10	'283 Patent - Claim 10	a mirroring driver of the local computer system <i>intercepting an operation on a selected data file associated with the file system performed by an application program executing on the local computer system, the operation representing a change to the selected data file; the mirroring driver comparing information identifying the selected data file with the configuration database and determining that the configuration database specifies that the selected data file is to be mirrored to the remote computer system;</i>
11 12 13	'283 Patent - Claim 16	a mirroring driver associated with the local computer system <i>for intercepting an operation performed by the at least one application program on a file included in the one or more data files, wherein the operation changes the file</i>

14

15 15. In my opinion, a person of skill in the art would find that each of the

16 “mirroring driver” and “mirroring driver means” limitations are expressed in functional

17 language as italicized in the chart above. In my opinion, a person of skill in the art would

18 find that the above claims do not recite sufficient structure to perform the recited functions

19 associated with the “mirroring driver” and the “mirroring driver means.” In particular, at the

20 time of the effective filing date of the asserted patents, the term “mirroring driver” in the

21 context of the asserted patents would not have been understood to have any defined meaning

22 to those of skill in the art. As such, the term “mirroring driver” does not carry its own

23 structure. I understand that because there is no structure in the “mirroring driver” and

24 “mirroring driver means” claim limitations to perform the recited functions, these claim

25 limitations should be interpreted as means-plus-function claim and are limited to the

26 corresponding structure disclosed in the patent to perform the recited functions.

27

28

1 16. The recited functions concerning the “mirroring driver” and “mirroring driver
2 means” vary somewhat from claim to claim. In each case, however, the “mirroring driver”
3 and “mirroring driver means” must, at a minimum, perform the function of intercepting only
4 change information or operations from selected files which are to be backed up.

5 17. In my opinion, a person of skill in the art would find that the corresponding
6 structure described in the specification to perform the recited functions is:

7
8 software configured to execute in an operating system
9 input/output stack which intercepts file changes and determines
 from a table which changes are to be backed-up

10 18. This structure is shown with reference to Figures 4A and 4B and is described
11 in the specification at column 2, lines 18-30 and at column 5, line 53 through column 7, line
12 39.

13 19. In my opinion, software which determines from a table which changes are to
14 be backed-up is needed so that only files on the local computer system that have been
15 selected to be mirrored are backed-up. This is dictated by the claim language that only
16 changes or operations from “selected” files are intercepted for backing-up and further
17 described in the specification which, for example, states in the “Summary of the Invention”
18 that “[t]he mirroring driver has a table with information read from the configuration database
19 and determines if the operation is on a protected file.” (Column 2, lines 27-29).

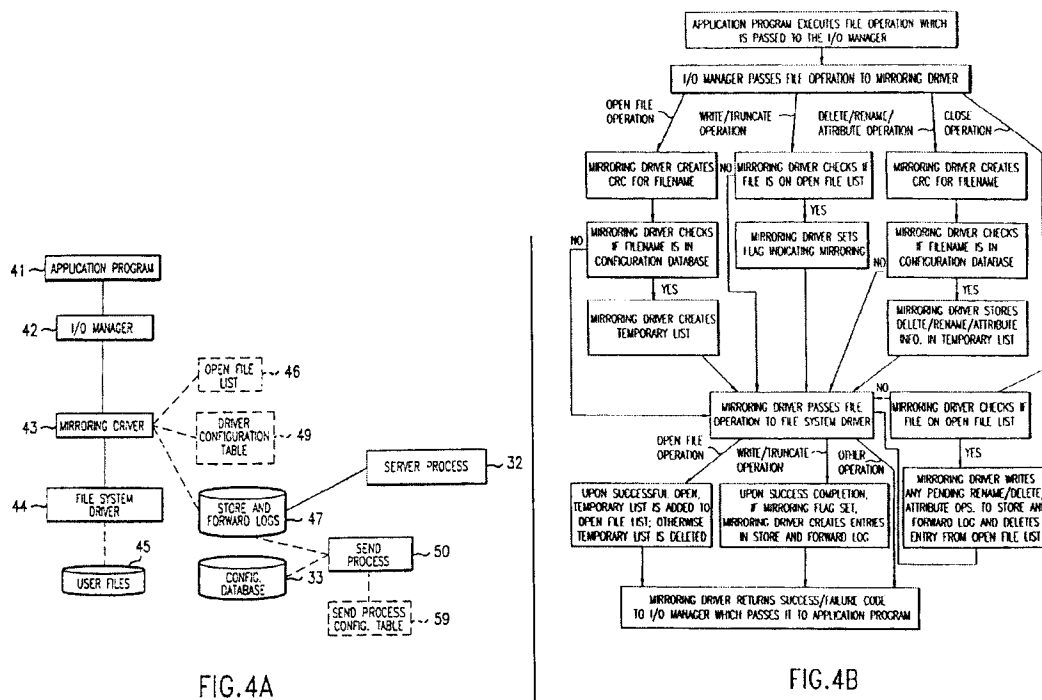
20 20. In my opinion, the software must also include software that is configured to
21 execute in an operating system input/output stack (i.e., to operate on data exchanged
22 between an I/O manager and a file system driver). This is needed for providing the function
23 of intercepting changes as they are being made to the files of the application program, as
24 provided by the claims. As explained in the specification:

25
26 File operations executed by application program 41 are passed
27 to the input-output (I/O) manager 42 of the local computer
28 system. I/O manager 42 passes the file operations to mirroring

driver 43, which in turn passes the file operation to file system driver 44.

(Col. 5, lines 56-60).

21. According to the specification, the “write intercept and store functions” (col. 2, lines 55-58) are illustrated in Figures 4A and 4B and demonstrate that the “mirroring driver” is configured to execute in an operating system input/output (I/O) stack on data exchanged between an I/O manager and a file system driver to perform the recited intercept function.



22. In my opinion, regardless of whether the “mirroring driver” and “mirroring driver means” limitations are expressed as means-plus-function elements, a person of skill in the art would understand these terms to mean “software executing in an operating system input/output stack which intercepts file changes and determines from a table which changes are to be backed-up.”

23. As explained above, it is my opinion that at the time of the effective filing date of the asserted patents, the term “mirroring driver” in the context of the asserted patents

1 would not have been understood to have any defined meaning to those of skill in the art. In
2 my opinion, a person of skill in the art would have to consult the specification and
3 prosecution history to determine what the meaning of “mirroring driver” is. Moreover, I am
4 not aware of any technical dictionary definitions for the term “mirroring driver” that would
5 be applicable to the asserted patents and I understand that neither of the parties to this case
6 have cited any such definitions for “mirroring driver.” In my opinion, a person of skill in the
7 art would find instead that a special definition of the term mirroring driver is provided by the
8 specification which, in my opinion, is accurately incorporated by the construction proposed
9 above.

10 24. Specifically, the “Summary Of The Invention” provides that “[t]he mirroring
11 driver has a table with information read from the configuration database and determines if the
12 operation is on a protected file.” (Col. 2, lines 27-29). Thus, in my opinion, a person of skill
13 in the art would understand that the mirroring driver includes software which determines
14 from a table which changes are to be backed-up. Additional support for this construction is
15 further found in the specification, for example, in Figures 3 and in Figures 4A and 4A
16 (illustrating the “write intercept and store function”) and the text describing the “Write
17 Intercept and Store” function beginning at column 5, line 53.

18 25. In my opinion, a person of skill in the art would also find the mirroring driver
19 defined as being configured to execute in the operating system input/output stack (i.e., to
20 operate on data exchanged between an I/O manager and a file system driver). In my opinion,
21 this construction is required by the claim language which recites that the mirroring driver
22 “intercepts” change information or operations on the local computer system. Moreover, as
23 discussed above, the “write intercept and store functions” shown and described in Figures 4A
24 and 4B provide for such a configuration.

25 26. Additionally, the specification expressly defines the mirroring driver as being
26 “attached” to the file system driver. For example, in the Summary Of The Invention, the
27 specification states that “[t]he system of the present invention provides a mirroring driver
28 that is *attached to the file system driver* of the local computer system and intercepts

operations on files.” (Col. 2, lines 18-21) (emphasis added). This aspect of the mirroring driver is described elsewhere in the specification. (See, e.g., abstract; col. 2. lines 23-27; col. 5, lines 63-65). As further explained below, “attaching” to the file system driver means that the mirroring driver resides as a software layer in the operating system input/output stack. Thus, the fact that the mirroring driver is expressly defined in the specification as being “attached” to “file system driver” further dictates that the “mirroring driver” be configured to execute in the operating system input/output stack (i.e., to operate on data exchanged between the I/O manager and a file system driver).

“mirroring driver is attached to the file system driver” and related terms

27. In my opinion, a person of ordinary skill would have understood the term “mirroring driver is attached to the file system driver” in the context of the asserted patents to mean that “the mirroring driver resides as a layer in the operating system input/output stack so that it can intercept data as it is exchanged between the I/O manager and the file system driver.”²

28. In my opinion, a person of ordinary skill in the art would understand that the term “attached” as used in the claims and specification of the asserted patent is a term of art, that is not accurately defined in a layman’s dictionary definition of “attach.” Specifically, “attaching” a driver to the file system driver in the context of the asserted patents would have meant to a person of skill in the art that the driver is inserted into a layer of the operating system input/output stack so that this “attached” driver could intercept data as it is exchanged between the I/O manager and the file system driver.

29. This meaning is fully supported by and consistent with the description of “attaching” found in “Chapter 11: The Layered Driver Mode” of my 1999 Windows NT Device Driver Development book:

² The terms “mirroring driver that is attached to a file system driver” in claim 9 of the ‘141 patent and “mirroring driver means is attached to the file system driver means” should be construed similarly according to the wording set forth in NSI’s Proposed Jury Instructions.

The Windows NT I/O Manager includes the capability for one Kernel mode driver to “attach” one of its Device Objects to a Device Object created by a different driver. The result of this is that IRPs [I/O Request Packets] *destined for the driver associated with the original Device Object will be sent to the driver associated with the “attached” Device Object.* This attached driver is a Filter driver. *The Filter driver can then examine, modify, complete, or pass along the IRPs it receives to the original driver. Filter drivers may be inserted at any layer in the Windows NT driver stack.*

(Exhibit D at p. 235-36) (NSI 027941-42)

30. The specification confirms that the inventors used the word “attach” as a term of art in accordance with the proposed construction above. The specification states that “[m]irroring driver 43 is attached to file system driver 44 using, for example, the IoAttachDevice call of Windows NT” and that “[i]n this way, the mirroring function can be implemented without requiring recompilation of application or operating system programs.” (Col. 5, lines 56-60); (*See also*, col. 2, lines 23-27: “By attaching the mirroring driver to the file system driver, the system of the present invention can mirror files accessed by existing application programs, having no data protection code, without modification to the application programs.”).

31. Only by “attaching” the mirroring driver to the file system driver as such (i.e., as a layer in the operating system input/output stack) is the mirroring driver able to intercept changes to the data files of the application program as they are passed from the I/O manager to the file system driver (as required by the recited “intercept” function). Such “attachment” also allows the mirroring driver function to be implemented without requiring recompilation of application or operating system programs as emphasized in the specification and prosecution history. (*See, e.g.*, col. 2, lines 23-27; col. 5, lines 56-60; November 10, 1997 Amendment p. 6; August 5, 1996 Amendment, p. 4).

“selected data file” and related terms

32. In my opinion, one of ordinary skill would have understood the term “selected data file” in the context of the asserted patents to mean “a data file specified in a configuration database by the user to be backed-up.”³

33. First, the word “select” is defined as “chosen from a number or group by fitness or preference.” *See Webster’s Third New Int’l Dictionary* (1993) (Ex. E). Consistent with this definition and the specification, the “selected data file” is the file specified and chosen by the user to be backed-up. Moreover, the file chosen by the user to be backed-up is in a configuration database:

- a user initializes a configuration database that specifies the data files on a local system the user wishes to back-up, (i.e., mirror) and the network location of a remote computer system to contain the back-up files. (Col. 2, lines 15-18)
- Configuration database 33 specifies the files to be mirrored . . . (Col. 4, lines 8-9)

34. Because the only way to tell whether a data file has been specified and chosen by the user to be backed-up is through reference to information from the configuration database, the term “selected data file” would have been understood by a person of skill in the art to mean “a data file specified in a configuration database by the user to be backed-up.”

“intercepting an operation on a selected data file residing in a file system performed by an application program” and related terms

35. In my opinion, one of ordinary skill would have understood the term “intercepting an operation on a selected data file residing in a file system performed by an application program” in claim 9 of the ‘141 patent to mean “seizing or capturing an operation

³ The term “selected data file” is used in claim 9 of the ‘141 patent. The terms “selected subset of the one or more data files” and “selected one of the one or more data files”, appearing in claims 1 and 14 of the ‘141 patent respectively, and the terms “selected file”, “selected data file”, and “one or more selected data files”, appearing in claims 1, 10 and 16 of the ‘283 patent respectively, should be construed similarly to “selected data file” according to the wording set forth in NSI’s proposed jury instructions.

1 on a selected file as it is being directed from the application program to the file system driver
2 in the operating system input/output stack.”⁴

3 36. This construction is consistent with the claim language requiring
4 “intercepting” with respect to a operation on a selected data file residing in a file system
5 performed by an application program and the ordinary meaning of the word intercept:

- 6 • **intercept** 1: to take, seize, or stop by the way or before arrival at the destined
7 place . . . (Webster’s Third New Int’l Dictionary) (1993), Exhibit E at NSI
8 038080)

9 37. According to the claims, the mirroring driver “intercepts” operations (or
10 change information) on selected data files performed (or made) by the application program.
11 In my opinion, a person of skill in art would understand that operations or changes made to
12 data files by an application program are applied to the data files by a file system driver. The
13 claims and specification confirm this. (See, e.g., Figs. 4A and 4B; col. 6, lines, 48-50).
14 Thus, in my opinion, based on the ordinary meaning of “intercept” and the claim language
15 concerning the intercept function, the mirroring driver seizes or captures operations or
16 changes which originate from the application program before arrival at the destined place
17 (i.e., the file system driver) in the operating system input/output stack. This is further
18 confirmed by the specification which illustrates the “write intercept and store function” in
19 Figures 4A and 4B (col. 2, lines 55-58) and the textual description of the “Write Intercept
20 and Store” function beginning at column 5, line 53.

21 38. The intercept function must seize or capture the operations or changes to
22 selected files as they are being directed from the application program to the file system driver
23 so that only operations or changes from selected files are backed-up at the remote computer
24 system. This construction is dictated by the claim language which provides that only
25 operations or changes to “selected” files are intercepted and backed-up at the remote system
26 and is consistent with the specification which explains:

27
28 ⁴ The “intercepting” limitations appearing in claim 14 of the ‘141 and claims 1, 10, and 16 should be
construed similarly according to the wording set forth in NSI’s proposed jury instructions.

- 1 • The mirroring driver has a table with information read from the configuration database and determines if the operation is on a protected file. If it is, the mirroring driver stores information regarding the operation in a log file. (Col. 2, lines 27-30)
- 2
- 3 • A first data protection program intercepts each write request, sends the write request to the disk drive controller and, for write requests to selected data files, locally stores a copy of the request. (col. 3, lines 17-22)
- 4

5 39. In addition, in my opinion, one of ordinary skill would have understood the
6 limitation that the mirroring driver “captures change information representing an individual
7 change to a file from a selected subset of the one or more data files by the at least one
8 application program” in claim 1 of the ‘141 patent to similarly mean that the mirroring driver
9 “seizes or captures an operation on a selected file as it is being directed from the application
10 program to the file system driver in the operating system I/O stack” for the reasons stated
11 above. Although this limitation does not expressly use the word “intercept”, I understand
12 that Legato, in its own construction of mirroring driver acknowledges that it must, at a
13 minimum, “intercept” changes to the data files. The above construction is further dictated by
14 the claim language and specification indicating that the mirroring driver is attached to the file
15 system driver. (*See, e.g.*, column 2, lines 18-23: “The system of the present invention
16 provides a mirroring driver that is attached to file system driver of the local computer system
17 and intercepts operations on files ...”). Thus, this limitation would have been understood to
18 have such a similar meaning.

19

20 ***“file system driver” and “file system driver means”***

21 40. In my opinion, one of ordinary skill would have understood the terms “file
22 system driver” and “file system driver means” in the context of the asserted patents to mean
23 “a software layer executing within an operating system’s input/output (I/O) stack to provide
24 an interface by which an I/O manager can perform I/O independent of underlying physical
25 devices.”

26 41. As recited in the claims and explained in the specification, the file system
27 driver “applies individual changes” to the data files that are made by the application program.
28 (*See, e.g.*, col. 6, lines 48-50). A person of ordinary skill in the art would understand that in

order to receive such individual changes and apply them to the data files, the file system driver resides in an operating system's input/output stack. This is confirmed in the specification, for example, in Figure 4A, which shows the "file system driver" in the input/output stack thereby allowing changes made by the application program 41 that are passed to the I/O manager 42 of that stack to reach the file system driver 44, which can then apply the actual changes to the user files 45.

42. The file system driver further provides an input/output interface allowing the I/O manager to perform I/O independent of the underlying physical device (e.g., the disk drive), consistent with its technical definition:

- File system driver A file system driver handles I/O independent of any underlying physical devices. File system drivers include drivers for the system-supplied NTFS and file allocation table (FAT) file systems. On the NT-based operating systems, file system drivers are kernel-mode drivers . . . *Microsoft Developers Network (MSDN) Library, Understanding Driver and Operating System Basics* (2003) (Exhibit F) (NSI 027886)

43. In addition, I understand that Legato has proposed to construe the term "file system driver" to mean "software that receives and/or invokes operations on data files in a file system." I understand that support for this construction is not based on any dictionary definition of "file system driver" but on definitions of the terms "file system" and "driver" in isolation. In my opinion, a person of skill in the art would not define "file system driver" as Legato proposes. Indeed, its definition could read on various types of software, including even application programs, which a person of skill in the art would understand are not "file system drivers" as further confirmed by the specification and claims which refer to these terms as separate elements.

"means for transmitting . . ."

44. I understand that the parties agree that the "means for transmitting . . ." limitations of claim 14 of the '141 patent and claim 16 of the '283 patent⁵ are expressed in

⁵ Claim 14 of the '141 patent recites a "means for transmitting information regarding each intercepted operation from the local computer system across the network to the remote computer system" and claim 16 of the '283 patent recites a "means for transmitting information regarding each intercepted operation from

means-plus-function format and are therefore limited to the corresponding structure described in the specification for performing the recited functions and equivalents thereof. In my opinion, one of ordinary skill would have understood the corresponding structure of the “means for transmitting . . .” limitations to be “a transmitting program configured to forward entries created by the mirroring driver in a store and forward log to the remote computer system.”

45. In my opinion, the corresponding structure utilized to perform the recited “transmitting functions” is illustrated generally in Figure 5 and described in the accompanying text (cited below) which discloses that send process 50, in conjunction with the store and forward logs 47 created by the mirroring driver with respect to selected files that are to be backed up, are utilized to transmit information to the remote computer system.

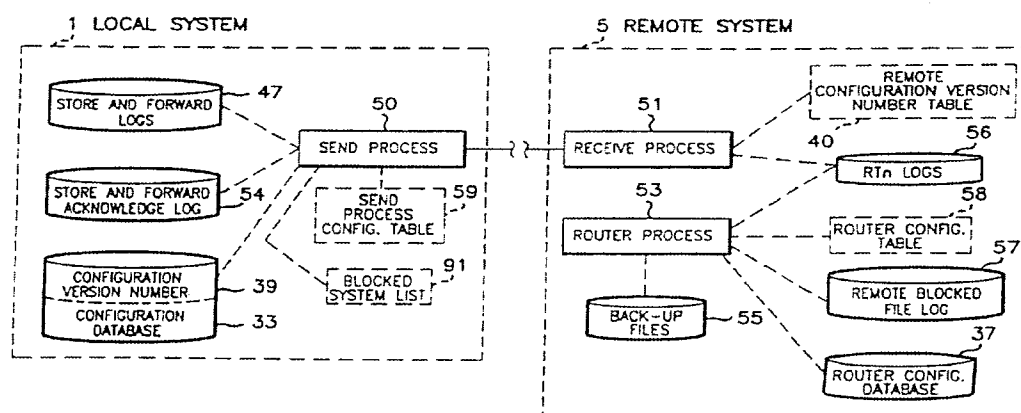


FIG. 5

If mirroring driver 43 receives a success code from file system driver 44 and the mirroring flag is set, *mirroring driver 43 creates one or more entries in Store and Forward Log 47*. As illustrated in FIG. 7, each entry in Store and Forward Log 47 comprises a source filename field 71, a CRC field 72, a command code field 73, a file offset field 74, a data size field 75, and a data field 76 . . . (Col. 6, lines 51-57)

Send process 50 executes in the background (i.e., asynchronously from other software on the local computer system) and periodically reads Store and Forward Logs 47 . . . (Col. 7, lines 49-51)

Once send process 50 locates a Store and Forward Log entry to read, send process 50 extracts the source filename and CRC

the local computer system across the network in the remote computer system if the file is included in the one or more files referenced in the configuration database.”

information from the entry. Send process 50 then scans the entire send process configuration table 59 and locates the entry with the matching CRC, preferably using a binary tree search algorithm. If the command code in the Store and Forward Log entry is other than a copy command (which is discussed below), *send process 50 then sends the source file, CRC, command code, offset, size and data fields of the Store and Forward Log entry, along with the current configuration version number 39, to the destination site specified in the send process configuration table entry . . .* (Col. 8, lines 22-33).

46. Thus, the corresponding “means for transmitting . . .” described in the specification is “a transmitting program configured to forward entries created by the mirroring driver in a store and forward log to the remote computer system.” The transmitting program forwards information in the store and forward logs, which the mirroring driver creates with respect to selected files, so that only operations to selected files are backed up at the remote computer system as required by the claims, the specification, and file history of the asserted patents.

“means associated with the remote computer system for updating a back-up copy . . .”

47. I understand that the parties agree that the “means associated with the remote computer system for updating a back-up copy . . .” limitations of claim 14 of the ‘141 patent and claim 16 of the ‘283 patent⁶ are expressed in means-plus-function format and are therefore limited to the corresponding structure described in the specification for performing the recited functions and equivalents thereof. In my opinion one of ordinary skill would have understood the corresponding structure of the “means for transmitting . . .” limitations to be:

programs executing on the remote computer system configured to receive change information sent by the means for transmitting, store the change information in a log file at the remote computer system and apply the change information to a back-up copy stored at the remote computer system.

⁶ Claim 14 of the ‘141 patent and claim 16 of the ‘283 patent each recite “means associated with the remote computer system for updating a back-up copy corresponding to the at least one of the one or more data files based on the transmitted information.”

1 48. In my opinion, a person of skill in the art would find that the corresponding
2 structure is shown generally in Figure 5 and is described from column 8, line 40 through
3 column 9, line 35, which discloses a receive process for receiving information and storing it
4 in a log file at the remote computer system (col. 8, lines 40-42) and which further discloses a
5 router process for applying the file update information to the back-up files (col. 9, line 11-
6 12).

7
8 ***“log file”***

9 49. In my opinion, a person of ordinary skill would have understood the term “log
10 file” appearing in claims 5 and 12 of the ‘141 patent and in claims 5 and 14 of the ‘283 patent
11 to mean “a file that lists the activities of a machine or program or users on a computer
12 system.” The term “log file” is a term of art which ordinarily is defined as “[a] file that lists
13 the activities of a machine or program or users on a computer system.” *See* Dictionary of
14 Computer and Internet Words (2001) (Ex. G). Based on my review of the patent and file
15 history, a person of skill in the art would give the term “log file” its ordinary meaning as
16 defined above.

17
18 ***“local computer system” and “remote computer system”***

19 50. In my opinion, a person of ordinary skill would have understood the term
20 “local computer system” appearing in the independent claims of the asserted patents to mean
21 “a system containing a computer and which can store data files that are accessible by the
22 application program.” This construction is consistent with the ordinary meaning a person of
23 skill in the art would attribute to a local computer system and the claim language which
24 associates the local computer system (as opposed to the remote system) as the system having
25 one or more data files in a file system that are accessed by at least one application program.

26 51. In my opinion, a person of skill in the art would have understood the term
27 “remote computer system” in the context of the asserted patents to mean “a computer system
28 that is separated by a distance greater than usual from a related, but more substantial

computer system.” This construction is consistent with the ordinary meaning that a person of skill in the art would attribute to a remote computer system and the definition of “remote” which is defined in the art as “[a] system or device that is separated by a distance greater than usual from a related but more substantial system or device . . .” *See Newton’s Telecom Dictionary* (2003) (Ex. H).

“file system”

52. In my opinion, a person of ordinary skill would have understood the term “file system” in the context of the asserted patents to mean “a combination of software and disk sector configuration that provides organizational structure for user’s files and manages the translation between the structure of files presented to the application program and the actual organization of those files on the disk.” This construction is consistent with the ordinary meaning that a person of skill in the art would attribute to the term “file system” and its definition in the art:

file system A combination of software and disk sector configuration that provides an organizational structure for a user’s files. A file system such as FAT organizes the files on disk and manages the translation between the structure of the files that the user sees on screen and the actual organization of the same files on disk . . . *See Dictionary of Computer and Internet Words* (2001) (Ex G) (NSI 027896)

“application program having no data protection code”

53. In my opinion, one of ordinary skill would have understood the term “application program having no data protection code” in the context of the asserted patents to mean “a computer program used for an application having no code for the protection of data.” This construction is consistent with the ordinary meaning of an application program and the claim language indicating that such program has data protection code.

“network”

54. In my opinion, a person of ordinary skill would have understood the term “network” in the context of the asserted patents to mean “a system of computers, and often peripherals such as printers, linked together.” The term “network” is a term of art which ordinarily is defined as “a system of computers, and often peripherals such as printers, linked together . . .” *See* Dictionary of Computer and Internet Words (2001) (Ex. G). Based on my review of the patent and file history, a person of skill in the art would give the term “network” its ordinary meaning as defined above

“acknowledgement message”

55. In my opinion, a person of ordinary skill would have understood the term “acknowledgement message” appearing in claims 6 and 13 of the ‘141 patent and in claims 7 and 15 of the ‘283 patent to mean “the transmission of a message from a receiving device to the sending device indicating that the data sent has been received correctly” The term “acknowledgement” is defined as “[i]n data communications, the transmission of acknowledgment (ACK) characters from the receiving device to the sending device indicates that data sent has been received correctly.” *See* Newton’s Telecom Dictionary (1995) (Exhibit I). Based on my review of the patent and file history, a person of skill in the art would give the term “acknowledgment message” its ordinary meaning as defined above

“configuration database”

56. In my opinion, a person of ordinary skill would have understood the term “configuration database” in the context of the asserted patents to mean “a database containing identities of the data files that the user chooses to be backed up.” This construction is consistent with what a person of ordinary skill in the art would discern from the term “configuration database” and its description in the specification:

- Configuration database 33 specifies the files to be mirrored. (Col. 4, lines 7-8).
- [A] user initializes a configuration database that specifies the data files on a local system the user wishes to back-up, (i.e. mirror) and the network location of a remote computer system to contain the back-up files. (Col. 2, lines 15-18).

1 57. I understand that Legato asserts that claim construction is needed for the
2 following additional terms: "data files", "change information", "individual change", "file"
3 and "change." In my opinion, no additional construction of these terms is necessary in light
4 of the claim constructions already provided and the claim language in which the terms
5 appear. To the extent that the Court does determine that these terms should be construed,
6 proposed constructions are set forth for "data files", "change information" and "file" in NSI's
7 proposed jury instructions, and in my opinion, those proposed constructions would be
8 consistent with the meaning a person of skill in the art would attribute to those terms in view
9 of claim language, the specification and the file history.

10
11 I declare under penalty of perjury that the foregoing is true and correct.

12
13 Executed this 14th day of May 2004

14 By: W. Anthony Mason
15 W. Anthony Mason
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